



THE ARCH: RAG & AGENTIC AI HACKATHON · LEGAL SERVICES TRACK

ComplianceMind

AI-Powered Legal Compliance for Indian SMEs

A grounded, citation-first compliance assistant — built on Retrieval-Augmented Generation, not guesswork.

The Problem Every SME Faces

Legal Blind Spots

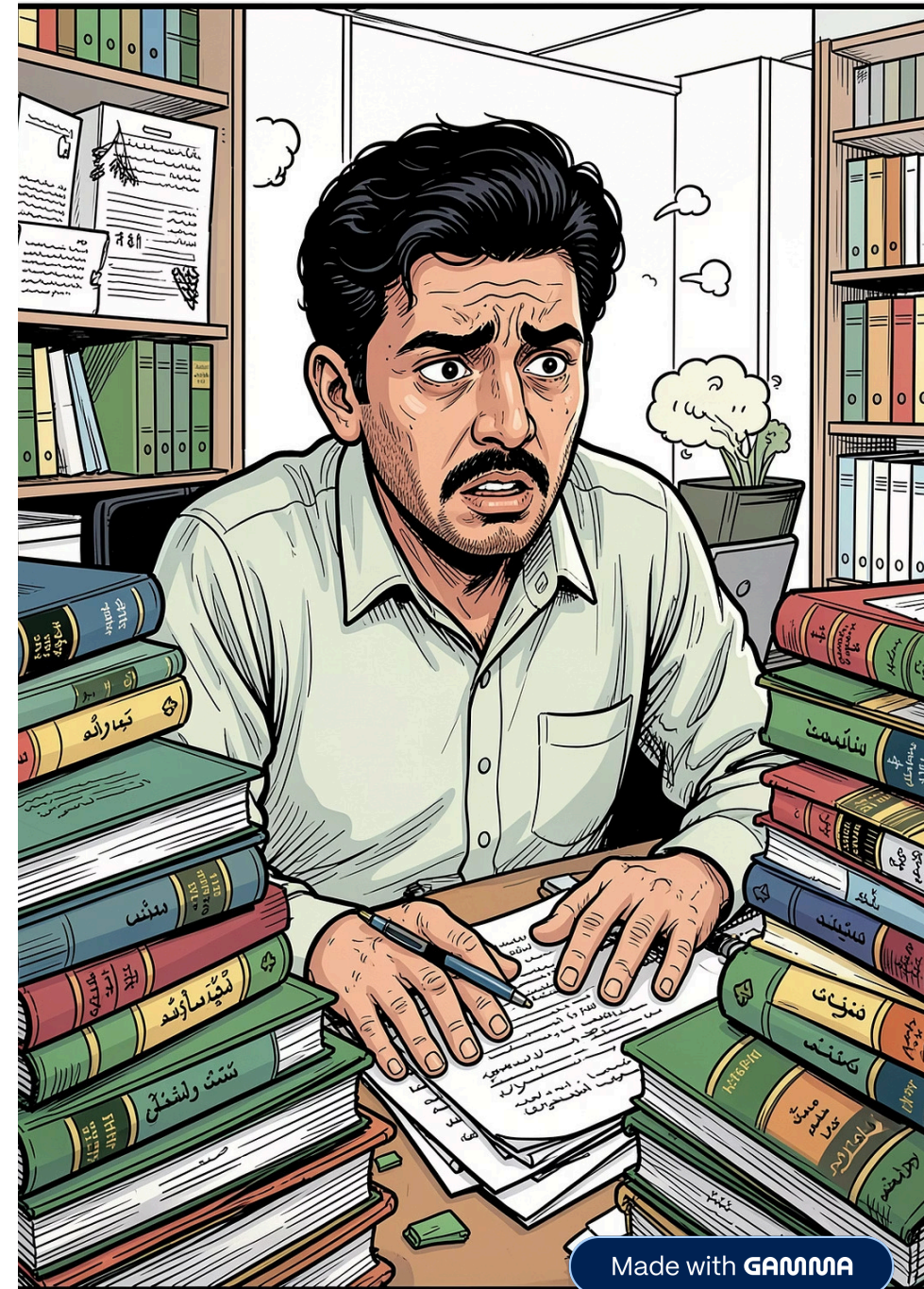
Indian SMEs don't know which laws govern their operations — data collection, e-commerce, payments, and more.

Hallucinated Advice

Generic LLMs paraphrase legal guidance without grounding — dangerous and potentially harmful in compliance contexts.

The Real Need

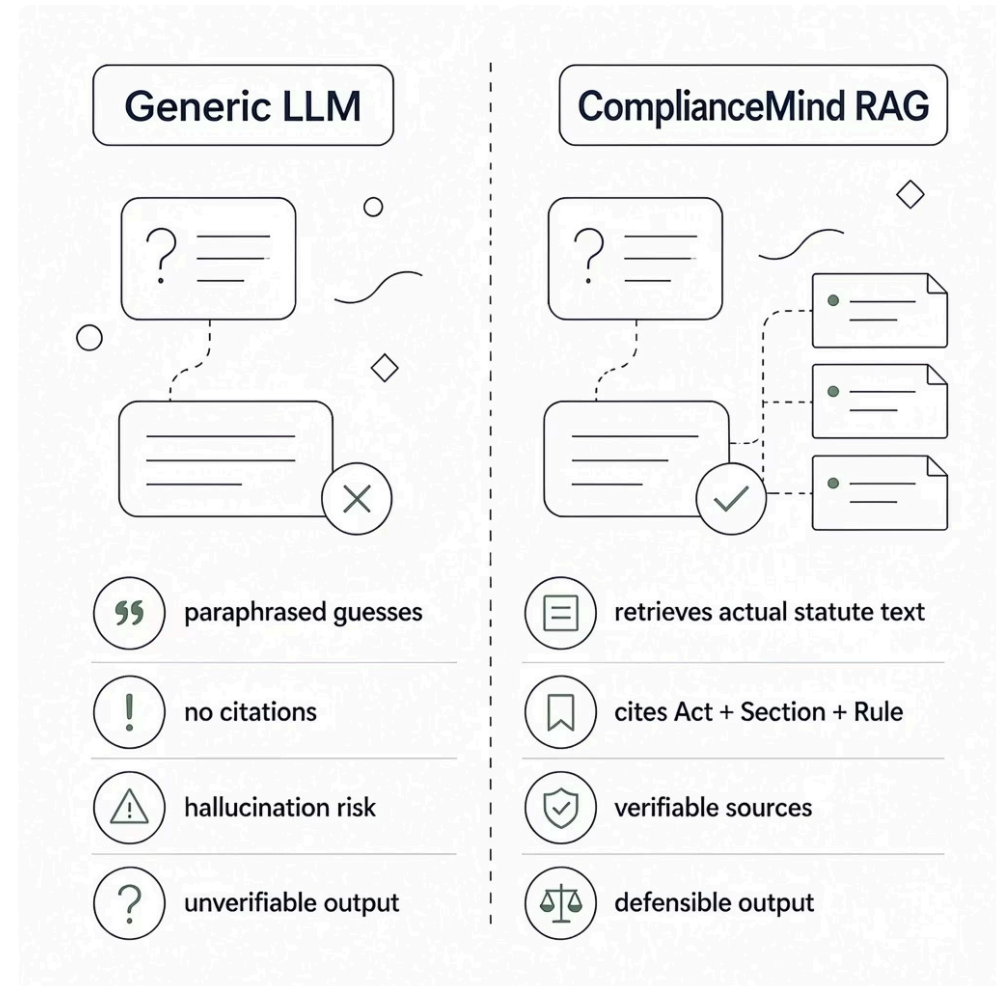
A system that retrieves **actual statute text** — not vague summaries or confident guesses.



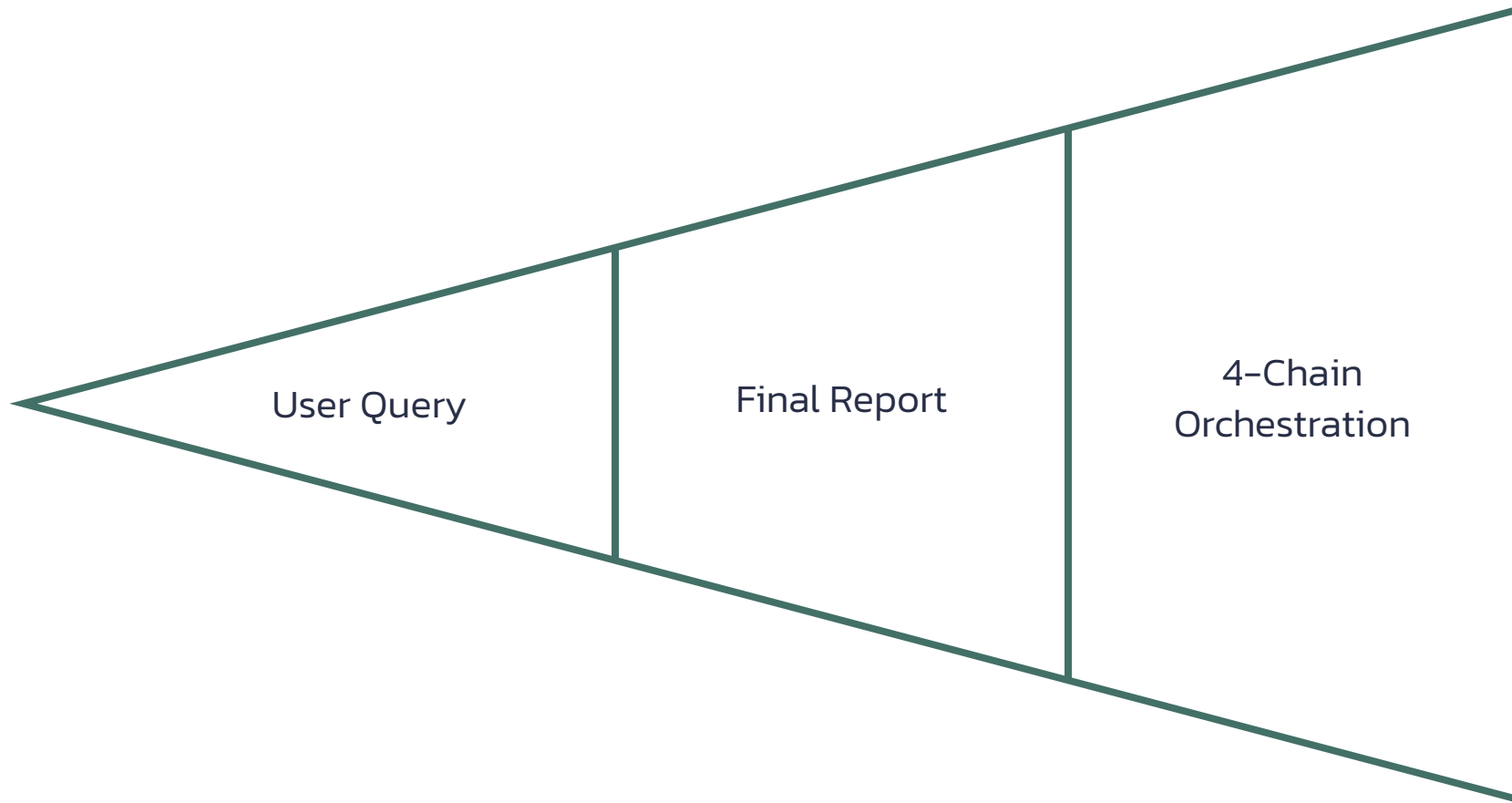
Why ComplianceMind Is RAG — Not "Just ChatGPT"

Every legal claim ComplianceMind makes is **grounded in retrieved, verifiable source text** from Indian statute law. The system cites the exact **Act, Section, and Rule** — not a vague summary.

Judges and reviewers can ask: *"Where did that come from?"* — and ComplianceMind points to an exact clause.



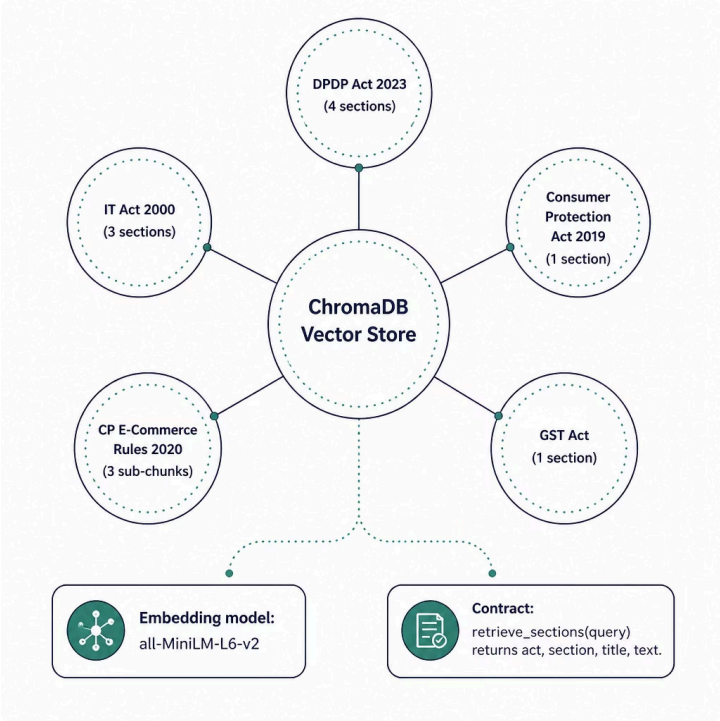
System Architecture



A clean 4-chain orchestration ensures every stage's output feeds the next — from raw query to a fully cited compliance report.

Knowledge Base: Grounded in Real Law

ComplianceMind's corpus covers **12 statute sections** sourced directly from primary government sources — **indiacode.nic.in** and official PDFs.



DPDP Act 2023
Sec 4, 5, 6, 8

GST Act
Sec 22

IT Act 2000
Sec 43A, 72, 72A

Consumer Protection Act 2019
Sec 2(47)

CP (E-Commerce) Rules 2020
Rule 4 (3 sub-chunks)

Engineering Challenges & How We Solved Them

1

False Positives from Large Chunks

Problem: Multi-topic Rule 4 chunks matched unrelated queries.

Fix: Split into 3 focused sub-chunks — improved retrieval precision.

2

Stale Embeddings in ChromaDB

Problem: "Ghost" embeddings persisted after corpus edits.

Fix: Rebuilt the vector store from scratch after structural changes.

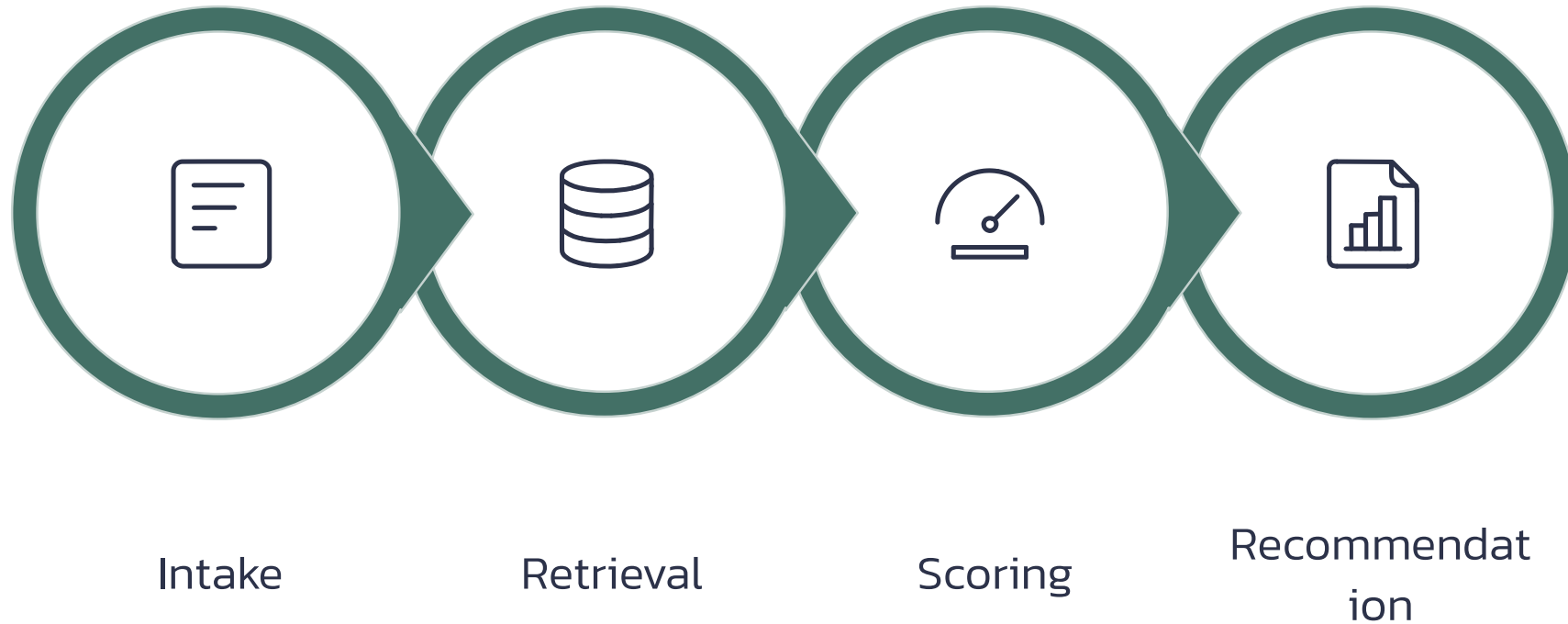
3

Sparse Corpus & Negative Queries

Problem: Offline business queries returned weak matches.

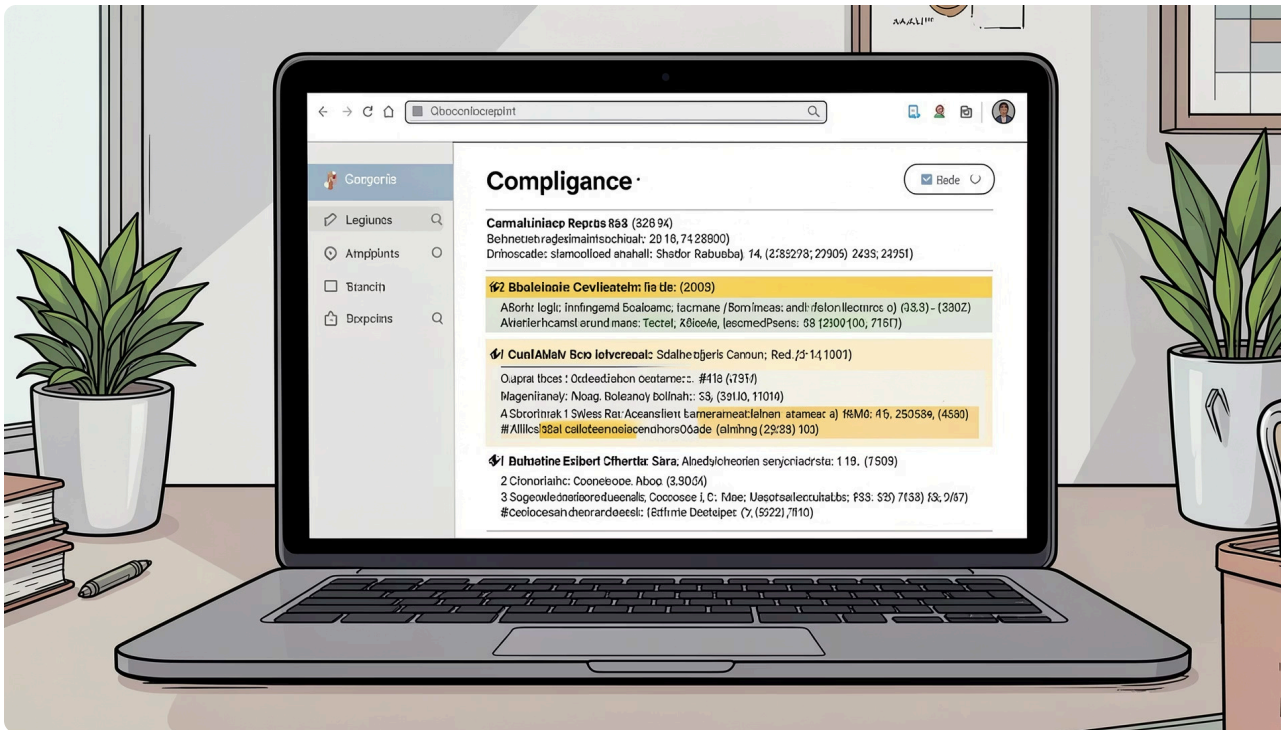
Fix: Pushed relevance judgment into the LLM reasoning layer — not just retrieval.

The 4-Chain Orchestration Pipeline



Each stage's structured output feeds the next — ensuring the final report is grounded, scored, and actionable. **LLM provider: Groq API** for low-latency inference.

Demo: From Query to Cited Report



Sample User Query

"I run an e-commerce store and collect customer phone numbers. What laws apply to me?"

ComplianceMind Output

- DPDP Act 2023 — Sec 5 (notice & consent obligations)
- CP (E-Commerce) Rules 2020 — Rule 4(2) (grievance officer)
- IT Act 2000 — Sec 43A (data breach liability)

 Every claim links to an exact Act + Section — fully auditable.

Tech Stack



ChromaDB

Persistent local vector store for statute embeddings using all-MiniLM-L6-v2.



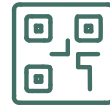
Groq API

Ultra-low-latency LLM inference powering all 4 orchestration chain stages.



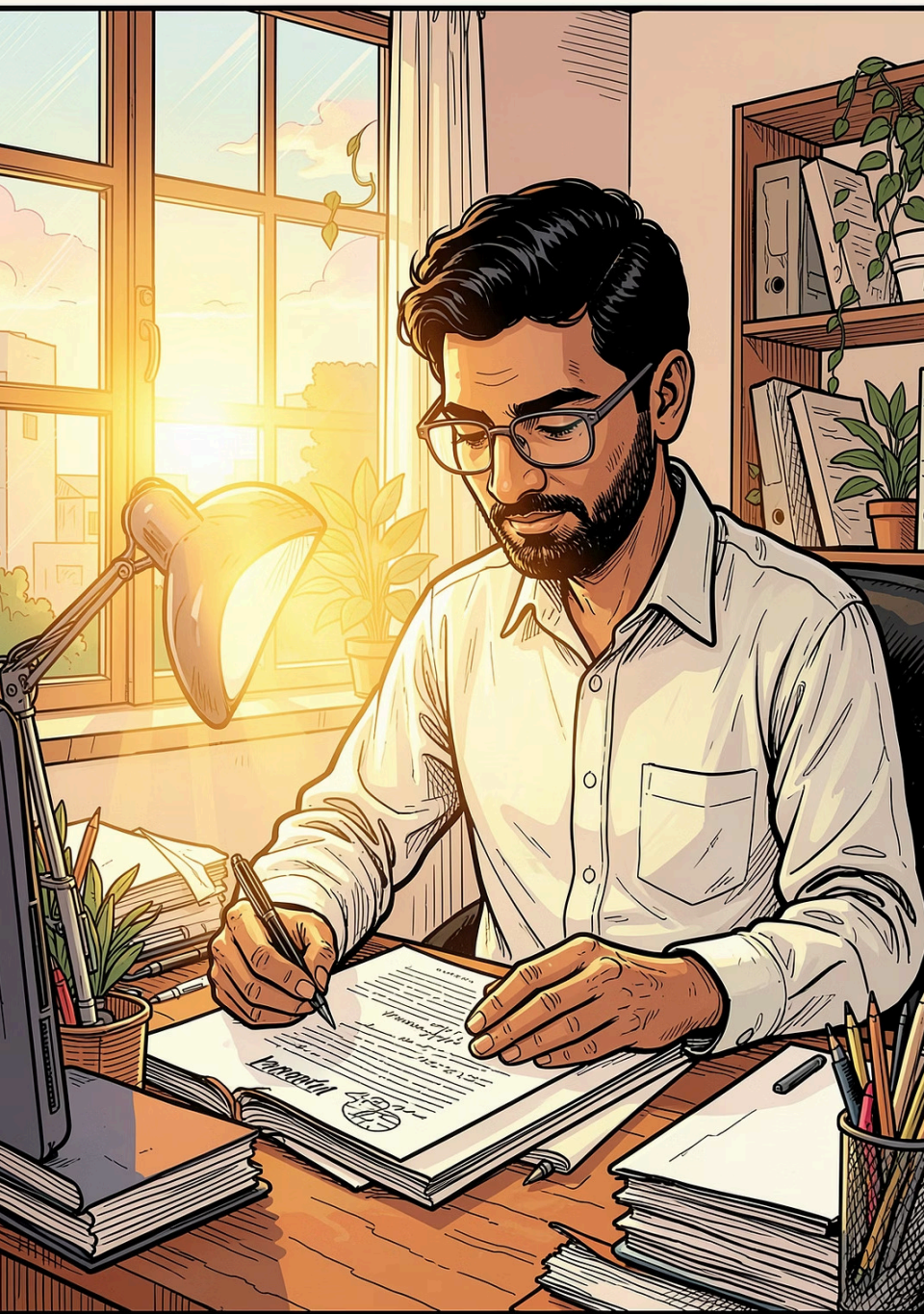
Streamlit

Lightweight Python UI (app.py) for query input and report display.



Python

Full backend in Python — pipeline.py orchestrates retrieval, scoring, and generation.



Impact & The Road Ahead

Why It Matters

ComplianceMind helps non-lawyers self-assess compliance risk in plain language — grounded in real statute law, making it **defensible, auditable, and trustworthy.**

Known Limitations

The 12-section corpus means sparse edge-case coverage. Mitigated by downstream LLM reasoning rather than blind retrieval trust.

What's Next

Expand corpus coverage, add re-ranking, and surface confidence scores — making ComplianceMind production-ready for Indian SMEs.